Steven Durr

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Machine learning researcher with a Ph.D. in physics. Transformers and sequence models, reinforcement learning, interpretability, and the dynamics of training.

Experience

Lead Scientist — STR, Boston, MA · May 2023 – Present (Lead Scientist since Oct 2026; Senior Scientist before)

- Lead deep-learning research for signal extraction and classification on complex time-series data; built the team's current best-performing model.
- Developed semi-supervised methods that improve generalization under sparse labels; ran large-scale architecture and hyperparameter searches.
- Set research direction for two researchers; advise program managers and domain experts on feasibility and model limitations.

Research Fellow — Supervised Program for Alignment Research (SPAR), Remote · Feb – Aug 2025

- Mechanistic interpretability with Andy Arditi on DeepSeek-R1-Distill-Llama-8B: how a reasoning model switches between thinking and answering.
- Found reasoning- and answering-mode activations linearly separable at nearly all layers; projection onto a difference-in-means direction flips sharply at the `</think>` boundary, an abrupt mode transition.
- Tested causality with single- and multi-layer activation steering, scored by a log-probability gap-closure metric on held-out prompts; a token-normalization control ruled out a token-frequency artifact.
- A single attention head that appeared to control the shift on short prompts did not survive a broader prompt distribution; reported it as a negative result. Writeup

Software Engineering Intern, Quantum AI — Google, Venice, CA · Summer 2021

- Ran surface-code calibration experiments on superconducting processors: inferred effective Hamiltonian parameters via Floquet resonance and compared alternative pulse sequences.

Software Engineering Intern, Google My Business — Google, Mountain View, CA · Summer 2018

- Clustered noisy business-website text with transformer- and LSTM-based (ELMo) representations to find content categories; built templated generation on them after finding direct neural generation too unreliable for production.

Graduate Researcher, Machine Learning and Physics — UCLA · Sep 2016 – Jun 2022

- **GAN training dynamics:** analytical and computational framework treating adversarial training as a dynamical system; characterized when mode collapse occurs versus convergence to the full distribution, validated on MNIST. First author, *Physical Review X* (2023), with IBM Research.
- **Transformer quantum states:** autoregressive transformers for many-body ground states, trained on self-generated samples with REINFORCE-style policy gradients; lower variational energies than DMRG on highly entangled systems. Diagnosed entropy collapse as the main failure mode; fixed with symmetry constraints and entropy regularization. Code
- **Stochastic optimal control:** GPU-accelerated Hamilton–Jacobi–Bellman solver (NumPy/CuPy) with memory-aware batching, Monte Carlo validation and an automated test suite. Code
- **Phase classification:** compared deep supervised nets and autoencoders with density-based and spectral clustering; the simplest method won and was published. First author, *Physical Review B* (2019).

Education

- **Ph.D., Physics — UCLA**, 2022. GPA 4.00/4.00. Dissertation: *Aspects of Many-Body Physics and Machine Learning*. (M.S., Physics, 2017)
- **B.A., Mathematics and Physics — Cornell University**, 2016. Mathematical Physics concentration; Dean's List.
- **Deep Learning Theory Summer School — Princeton**, 2021.

Publications

- **S. Durr**, Y. Mroueh, Y. Tu, S. Wang. "Effective Dynamics of Generative Adversarial Networks." *Physical Review X* 13, 041004 (2023).
- I. E. Powell, **S. Durr**, N. Rombes, S. Chakravarty. "Density Wave Mediated Dzyaloshinskii–Moriya Interactions." *Physical Review B* 103, 024433 (2021).
- **S. Durr**, S. Chakravarty. "Unsupervised Learning Eigenstate Phases of Matter." *Physical Review B* 100, 075102 (2019).

Skills

- **Machine learning:** transformers and sequence modeling, generative and autoregressive models, self-supervised and semi-supervised learning, reinforcement learning, GANs, energy-based models.
- **Interpretability:** linear probes, difference-in-means directions, activation steering, attention-head patching, sparse autoencoders, NNsight.
- **Engineering:** Python (10+ years), PyTorch, TensorFlow, NumPy, SciPy, CuPy, Optuna, Hydra, Git, HDF5; GPU and cluster workflows, automated testing.
- **Mathematics:** stochastic processes, Langevin and Fokker–Planck dynamics, statistical mechanics, optimal control and dynamic programming, Monte Carlo methods, information theory.